Pre-Analysis Plan for Testing Hypotheses Using SOEP Data

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Hypothesis 1 (H1): Having daughters leads to stronger support for gender equality (as measured by attitudes).

(1) Variables

- - **Dependent Variable (X)**: Attitudinal support for gender equality (e.g., `famw_egal` from SOEP, where available, measured on a Likert scale).

- Recode such that higher values indicate greater support.

- - **Independent Variable**: Indicator for having at least one daughter (binary: 1 = has daughter(s), 0 = no daughters).

- Use child-level data (`biokind`, `bioparen`) to construct household-child gender composition.

- - **Covariates**: Age, education, income, number of children, employment status, year fixed effects, region dummies.

(2) Inclusion/Exclusion Criteria

- - Include respondents with at least one biological child (using `bioparen`).
- - Limit analysis to survey waves from 2000–2020, as defined in project instructions.
- - Exclude respondents with missing gender information for children.

(3) Exact Analysis

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regress gender_equality_attitude has_daughter age educ income num_children employed i.year i.region, robust

- - **gender_equality_attitude**: recoded version of attitude variable.
- - **has_daughter**: binary treatment.
- - Use robust standard errors clustered at the household level.

(4) Interpretation of Results

- - Key coefficient: `has_daughter`.
- - A positive and statistically significant coefficient (p < 0.05) supports H1.
- - Report coefficient, robust SE, t-statistic, p-value, N, R^2.

Hypothesis 2 (H2): Having daughters leads to more left-leaning political preferences.

(1) Variables

- - **Dependent Variable (Y)**: Political self-placement (`plapol` or similar) on left-right spectrum (e.g., 0 = extreme left, 10 = extreme right).

- Reverse code to indicate higher values = more left-leaning.

- - **Independent Variable**: Same binary `has_daughter`.
- - **Covariates**: Age, gender, education, income, employment, household size, year and region fixed effects.

(2) Inclusion/Exclusion Criteria

- - Include only respondents with valid political self-placement.
- - Include only those with children.
- - Years 2000–2020.

(3) Exact Analysis

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regress left_leaning_politics has_daughter age gender educ income employed hhsize i.year i.region, robust

- - Use robust standard errors.

(4) Interpretation of Results

- - Coefficient on `has_daughter` indicates whether parents with daughters lean more left.
- - Hypothesis supported if coefficient is significantly positive ($p < 0.05$).

Hypothesis 3 (H3): Having daughters leads to more gender equality in the household (as measured by behaviors or outcomes).

(1) Variables

- - **Dependent Variable (Z)**: Share of household chores done by male respondent or indicator of equal task division (e.g., `dvhwork`, `dhhwork`, or time-use logs).

- Construct or recode as needed to range from 0 (traditional) to 1 (egalitarian).

- - **Independent Variable**: `has_daughter`.
- - **Covariates**: Age, education, number of children, income, employment status, fixed effects.

(2) Inclusion/Exclusion Criteria

- - Include households with heterosexual couples and valid time-use data.
- - Restrict to years with time-use module (2000–2020 where available).

(3) Exact Analysis

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regress household_gender_equality has_daughter age educ income num_children employed i.year i.region, robust

- - Use robust SEs, cluster by household.

(4) Interpretation of Results

- - Positive, significant coefficient on `has_daughter` supports H3.
- - Report effect size and infer directionality.

Hypothesis 4 (H4): Having daughters leads to a larger increase in the support of gender equality for men than for women.

(1) Variables

- - **Dependent Variable (X)**: Same as in H1.
- - **Key Independent Variables**:

- `has_daughter`

- `male`: binary gender indicator (1 = male)

- `has_daughter * male`: interaction term

- - **Covariates**: Age, education, income, employment, fixed effects.

(2) Inclusion/Exclusion Criteria

- - Include only respondents with known gender and children.
- - Limit years to 2000–2020.

(3) Exact Analysis

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regress gender_equality_attitude has_daughter male c.has_daughter#c.male age educ income employed i.year i.region, robust

- - Robust SEs; interaction term captures differential effect.

(4) Interpretation of Results

- - Focus on the interaction coefficient `has_daughter#male`.
- - Positive and significant value indicates stronger effect for men, confirming H4.

Each regression will be reported in a separate table with: coefficients, robust standard errors, test statistics, p-values, R^2, and N.

We will export results as CSV files as specified in the project instructions